

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Scenario-Based Questions

CO-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate	Shows correct relationships with minor gaps	Relationships are weak, unclear, or	No meaningful relationships established

		relationships between concepts		partially incorrect	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand